CHIRAG R. RAO

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Education

Johns Hopkins University — **Baltimore MD**
M.S. Computer Science — *June 2015–May 2018*

Cornell University — **Ithaca NY**
B.S. Electrical & Computer Engineering — *May 2009–Aug. 2013*

Awards & Leadership

- Received "Best Paper Award of the 2019 ICMCIS Conference"
- Received the Science, Mathematics, and Research for Transformation (SMART) full-tuition merit scholarship awarded by the US Department of Defense, 2010
- **Cornell University:**
  - President of Cornell University Eta Kappa Nu (HKN) Electrical Engineering Honor Society
  - Graduate of Cornell University with Latin Honors, *cum laude*
  - Dean's list, Cornell University, 2009-2013
  - Cornell University Joan and Irwin Jacobs Scholarship, 2009
  - Member of Cornell University Tau Beta Pi Engineering Honor Society

Experience

US Army Research Laboratory — **Adelphi MD**
Computer Engineer — *July 2013–present*
The US Army's corporate research laboratory, a $2B operation delivering long-term technological capabilities

- Developed a digital video streaming transmitter/receiver in GNU Radio to wirelessly transmit/receive data in the low-VHF RF band (Used to acquire results of an experiment published in IEEE VTC 2019)
- Developed a moving-target defense paradigm to make autonomic resource management tools robust for edge computing (in collaboration with UCLA and UMass, accepted for publication at MILCOM 2019)
- Simulated the re-use of Loosely-Synchronous codes to explore solutions to scaling dense, strongly-connected wireless networks implementing QS-CDMA (Published in IEEE PIMRC 2018)
- Developed a codebase to acquire and publish channel state information to the Robot Operating System from software-defined radios integrated into ground robots
- Conducted outdoor experimentation with ground robots to characterize the low-VHF RF band in near-ground, line-of-sight and non-line-of-sight communications in cluttered urban environments

Top Score Education — **Washington DC**
Professional Tutor — *Dec. 2018–present*
A personalized tutoring company that has helped over 600 students achieve their standardized testing goals since 2006

- Managed client relationships with students and parents, providing guidance and weekly progress reports
- Taught students mathematical concepts from Advanced Placement BC Calculus and the SAT exam
- Developed lesson plans and formulated test preparation recommendations for students and parents

Cornell University **Ithaca NY**
Academic Excellence Workshop Facilitator *Aug. 2011–Dec. 2012*
A College of Engineering initiative to provide students with optional supplementary math courses

- Delivered lectures to undergraduate engineers in semester-long supplementary math courses
- Organized class activities that encouraged students to learn the class material collaboratively

Activities & Community Service

Community Tax Aid, Inc. **Washington DC**
Tax Reviewer *Jan. 2014–Apr. 2019*
Prepared taxes and consulted free of charge for low-income households in the Washington DC metropolitan area

- Managed and advised volunteer tax preparers to ensure quality preparation of over 400 returns
- Prepared tax returns for over 60 households per year as an IRS Certified Advanced Tax Preparer
- Counseled clients in English and Spanish on tax-related topics and personal finance

South Asian Performing Arts Network **Washington DC**
Resident Dancer & Actor *Oct. 2018–present*
Performed classical and modern South Asian dance and theater pieces in the Washington DC metropolitan area

Languages

English/Kannada: Bilingual Native Proficiency

Spanish: Limited Working Proficiency

Computer skills

Analysis & Simulation: Matlab, Jupyter
Robotics: Robot Operating System, Rviz
Software-Defined Radio: GNU Radio, UHD
Python: Numpy, Tensorflow, Scikit-learn, Pandas
Development Tools: Bash, Docker, Git
Data Management: SQL, Hadoop, Spark

Publications

Refereed Journal Publications

F. T. Dagefu, G. Verma, **C. R. Rao**, L. Y. Paul, J. R. Fink, B. M. Sadler, and K. Sarabandi, "Short-range low-vhf channel characterization in cluttered environments," *IEEE Transactions on Antennas and Propagation*, vol. 63, no. 6, pp. 2719–2727, 2015.

Refereed Conference Publications

J. Choi, **C. R. Rao**, and F. T. Dagefu, "Real-Time Digital Video Streaming at Low-VHF for Compact Autonomous Agents in Complex Scenes," pp. 1–5, Institute of Electrical and Electronics Engineers (IEEE), 2019.

C. R. Rao, F. Dagefu, P. Spasojević, and G. Verma, "Node recruitment in Loosely-Synchronized scalable networks with sporadic transmission (poster)," in *2019 22nd International Conference on Information Fusion (FUSION) (FUSION 2019)*, (Ottawa, Canada), 2019.

J. Noor, A. Ali-Eldin, L. A. Garcia, **C. R. Rao**, V. Dasari, D. Ganesan, B. Jalaian, P. Shenoy, and M. B. Srivastava, "The case for robust adaptation: Autonomic resource management is a vulnerability," in *MILCOM 2019 Track 5 - Big Data and Machine Learning (MILCOM 2019 Track 5)*, (Norfolk, USA), 2019.

F. T. Dagefu, J. N. Twigg, **C. R. Rao**, and B. M. Sadler, "Directional communication enabled by mobile

parasitic elements," in *2019 International Conference on Military Communications and Information Systems* (*ICMCIS*), pp. 1–7, 2019.

C. R. Rao, F. T. Dagefu, G. Verma, P. Spasojević, and B. M. Sadler, "Scalable sporadic medium access for complex propagation environments," in *2018 IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications* (*PIMRC*), pp. 1–7, IEEE, 2018.

F. T. Dagefu, G. Verma, **C. R. Rao**, P. L. Yu, B. M. Sadler, and K. Sarabandi, "Measurement and characterization of the short-range low-VHF channel," in *2015 IEEE Wireless Communications and Networking Conference, WCNC 2015*, Institute of Electrical and Electronics Engineers Inc., 2015.

Technical Reports

F. T. Dagefu, G. Verma, **C. R. Rao**, P. L. Yu, J. R. Fink, B. M. Sadler, and K. Sarabandi, "Low vhf channel measurements and simulations in indoor and outdoor scenarios," Tech. Rep. ADA618985, U.S. Army Research Laboratory, 2015.

L. Sadler, **C. R. Rao**, J. Rogers, and H. Nguyen, "Rostojausbridge manual," Tech. Rep. ADA559379, U.S. Army Research Laboratory, 2012.